DENGUE

Custom Study Summary based on Government of India guidance

Overview

Dengue is a mosquito-borne viral disease caused by dengue viruses and transmitted mainly by infected *Aedes* mosquitoes. The illness can range from mild fever to severe dengue, which may involve plasma leakage, severe bleeding, shock and organ impairment.

Key Facts

• Causative agent: Dengue virus (DENV), with four serotypes: DENV-1, DENV-2, DENV-3 and DENV-4.

• Main vector: *Aedes aegypti* mosquito; other *Aedes* species may also transmit infection.

• Transmission: Primarily through the bite of an infected mosquito.

• Incubation period: Commonly about 4–10 days after an infected mosquito bite.

• Dengue may present as dengue fever, dengue with warning signs, or severe dengue.

• Early recognition of warning signs and appropriate fluid management are important in preventing complications.

Transmission

The usual transmission cycle is:

Infected person → Aedes mosquito → mosquito becomes infected → bite of another person → Dengue infection

• Aedes mosquitoes are primarily daytime biters, with increased activity during early morning and late afternoon.

• An infected person can serve as a source of virus for mosquitoes during the viraemic phase.

• Because the same vectors can transmit other arboviral diseases, dengue symptoms may overlap with illnesses such as chikungunya.

Clinical Features and Symptoms

Dengue should be suspected in a person with acute fever, especially in areas where dengue transmission occurs. Common manifestations may include:

• High fever

• Severe headache

• Pain behind the eyes

• Muscle and joint pain

• Nausea and vomiting

• Skin rash

• Fatigue and weakness

Warning Signs

Warning signs usually require prompt medical assessment and close monitoring. Important warning signs include:

• Severe abdominal pain or persistent vomiting

• Clinical fluid accumulation

• Mucosal bleeding

• Lethargy or restlessness

• Liver enlargement

• Laboratory evidence such as increasing haematocrit with rapidly decreasing platelet count

Severe Dengue

Severe dengue is a potentially life-threatening condition. It may involve:

• Severe plasma leakage leading to shock or respiratory distress

• Severe bleeding

• Severe organ involvement, including liver, central nervous system or other organ impairment

Diagnosis

Diagnosis is based on clinical assessment together with appropriate laboratory testing. Tests may include:

• Detection of dengue viral antigen such as NS1 during the early phase

• Molecular testing such as RT-PCR where available

• Dengue-specific IgM antibody testing

• Other supportive laboratory investigations including complete blood count and haematocrit monitoring

Treatment and Management

There is no specific antiviral treatment for dengue. Management is mainly supportive and depends on the clinical severity and phase of illness.

• Adequate oral fluids may be advised for patients who can tolerate oral intake.

• Paracetamol may be used for fever as advised by a healthcare professional.

• Aspirin and other NSAIDs should generally be avoided because of bleeding risk.

• Patients with warning signs or severe dengue require urgent medical evaluation and appropriate fluid management.

• Severe dengue may require hospital admission, careful monitoring and management of shock, bleeding and organ dysfunction.

Prevention and Control

Dengue prevention focuses on avoiding mosquito bites and reducing mosquito breeding sites.

Personal protection:

• Use mosquito repellents as appropriate.

• Wear clothing that reduces exposed skin.

• Use screens and other measures to prevent mosquito entry.

• Take precautions against mosquito bites during daytime as well.

Source reduction:

• Do not allow water to stagnate in containers.

• Empty, clean and cover water-storage containers regularly.

• Discard or manage objects that collect rainwater.

• Maintain environmental cleanliness and support local vector-control activities.

When to Seek Immediate Medical Attention

• Persistent vomiting or severe abdominal pain

• Bleeding from nose or gums, vomiting blood, or black stools

• Extreme weakness, restlessness or unusual drowsiness

• Difficulty breathing

• Cold or clammy skin, dizziness or fainting

• Reduced urine output or any rapid worsening after fever begins to subside

Dengue: Quick Summary

Cause: Dengue virus
Spread: Infected Aedes mosquito bites
Incubation: Commonly 4–10 days
Main symptoms: Fever, headache, body aches, nausea and rash
Warning signs: Abdominal pain, persistent vomiting, bleeding, lethargy and fluid accumulation
Diagnosis: Clinical assessment with appropriate laboratory testing
Treatment: Supportive care; urgent management for warning signs or severe dengue
Prevention: Mosquito-bite protection and elimination of breeding sites

Sources: National Guidelines for Clinical Management of Dengue Fever (2023), National Centre for Vector Borne Diseases Control (NCVBDC), Ministry of Health & Family Welfare, Government of India; NCVBDC Dengue Case Management guidance.

Educational summary only. This document is not a substitute for professional medical advice or emergency care.